

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	CH.Saisri	Reg. No.:	192472367
Section / Batch:	Cse-ai	Date:	

Code of Conduct

I ch.saisri (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

1. What is discourse in Natural Language Processing?
2. What is discourse analysis?
3. What is the interlingua approach in machine translation?
4. What is metaphor in natural language?
5. Analyze the transfer approach for translating between two languages.
6. Analyze why pronoun resolution is difficult in long documents.
7. How can text coherence be applied in automatic essay generation?
8. How can conversational agents be used in education?
9. How can dialog systems be applied in banking?
10. How can coherence checking be applied in grammar correction software?
11. How can discourse structure be used in question answering systems?
12. How can surface realization be applied to data-to-text systems?
13. A translation system analyzes English grammar before converting the sentence into Japanese grammar. Which translation approach is being used?
14. A chatbot interprets "Can you pass the salt?" as a question about ability. What is the problem?
15. A translation system is designed separately for English-to-French, English-to-German, and English-to-Spanish. What limitation can an interlingua approach solve?
16. A generated paragraph uses "therefore," "however," and "because" correctly. How do these improve the text?
17. What is a conversational agent?
18. What are the major components of a conversational agent?
19. What is Natural Language Generation?
20. What is the architecture of an NLG system?

ANSWERS

1.What is discourse in Natural Language Processing?

Answer:

Discourse refers to a sequence of related sentences that together communicate a meaningful idea. In NLP, discourse analysis studies how sentences are connected and how meaning is maintained across an entire text or conversation.

2.What is discourse analysis?

Answer:

Discourse analysis is the process of studying language beyond individual words and sentences. It examines relationships between sentences, speaker intentions, context, coherence, and the overall structure of communication.

3. What is the interlingua approach in machine translation?

Answer:

The interlingua approach represents the meaning of a source sentence using a language-independent intermediate representation. This representation can then be used to generate the sentence in different target languages.

4. What is metaphor in natural language?

Answer:

A metaphor is an expression in which one concept is understood in terms of another concept. Metaphor interpretation is challenging for NLP because the intended meaning may differ from the literal meaning.

5. Analyze the transfer approach for translating between two languages.

Answer:

The source sentence is first analyzed to identify its grammatical and semantic structure. This structure is transformed according to target-language rules before generating the final translated sentence.

6. Analyze why pronoun resolution is difficult in long documents.

Answer:

A pronoun may refer to an entity mentioned several sentences or paragraphs earlier. The system must track entities, grammatical features, semantic information, and discourse context to resolve the correct reference.

7. How can text coherence be applied in automatic essay generation?

Answer:

An essay generation system can organize ideas into introduction, body, and conclusion sections. It can also use transitions and discourse relations to ensure that paragraphs connect logically

8. How can conversational agents be used in education?

Answer:

Educational conversational agents can answer student questions, provide explanations, conduct quizzes, and offer learning guidance. They can also adapt their responses according to student interactions.

9. How can dialog systems be applied in banking?

Answer:

Banking chatbots can answer questions about balances, transactions, cards, and account services. Dialog management helps maintain the context of the customer's request throughout the conversation.

10. How can coherence checking be applied in grammar correction software?

Answer:

Grammar correction tools can analyze whether sentences logically connect with surrounding content. This allows the system to detect not only grammatical mistakes but also possible inconsistencies in meaning

11. How can discourse structure be used in question answering systems?

Answer:

The system can analyze which sentences contain explanations, causes, examples, or conclusions. This helps it locate the most relevant information when answering a user's question.

12. How can surface realization be applied to data-to-text systems?

Answer:

A data-to-text system first converts structured data into an internal representation of meaning. Surface realization then produces fluent sentences with proper grammar, word order, and vocabulary.

13. A translation system analyzes English grammar before converting the sentence into Japanese grammar. Which translation approach is being used?

Answer:

This represents a transfer-based machine translation approach. The system analyzes the source-language structure and transforms it according to the grammatical and structural rules of the target language.

14. A chatbot interprets "Can you pass the salt?" as a question about ability. What is the problem?

Answer:

The chatbot has failed to correctly interpret the dialog act and speaker intention. Although grammatically a question, the utterance functions as a polite request.

15. A translation system is designed separately for English-to-French, English-to-German, and English-to-Spanish. What limitation can an interlingua approach solve?

Answer:

Separate translation systems require many language-pair combinations as the number of languages increases. An interlingua can provide a shared semantic representation that simplifies multilingual translation

16. A generated paragraph uses "therefore," "however," and "because" correctly. How do these improve the text?

Answer:

These discourse markers explicitly show logical relationships such as cause, contrast, and conclusion. Their correct use improves cohesion and helps readers understand how ideas are connected.

17. What is a conversational agent?

Answer:

A conversational agent is a computer system that interacts with users using natural language. Examples include chatbots, virtual assistants, customer support bots, and voice assistants.

18. What are the major components of a conversational agent?

Answer:

A conversational agent generally contains language understanding, dialog management, knowledge processing, and response generation components. These components work together to understand users and generate meaningful replies.

19. What is Natural Language Generation?

Answer:

Natural Language Generation, or NLG, is the process of automatically producing humanreadable language from structured data or internal representations. It converts information into meaningful sentences, paragraphs, or complete documents.

20. What is the architecture of an NLG system?

Answer:

An NLG architecture generally includes content selection, discourse planning, sentence planning, lexical selection, and surface realization. These stages transform structured information into grammatically correct natural language.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	3	3	4	4	3	8	83
2	3	4	3	4	3	8	
3	3	3	4	4	3	8	
4	3	4	3	4	3	8	
5	3	3	4	4	3	9	
6	3	4	3	4	3	10	
7	3	3	4	4	3	8	
8	3	4	3	4	3	8	
9	3	3	4	4	3	8	
10	3	4	3	4	3	8	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____